

Module 4: Understanding processes

This module provides progression from the application-oriented work in Physics in Action. Understanding Processes is organised around different ways of describing and understanding processes of change: motion in space and time, wave motion, quantum behaviour. It provides a sound foundation in the classical physics of mechanics and waves and takes the story further, touching on the quantum probabilistic view.

Physical variables are extended from scalars to quantities that add like vectors.

Either section may be covered first. Some teachers may wish to introduce work on vector addition from **4.2** before embarking on work on combining phasors.

The first section of the module is mainly about superposition phenomena of waves with a brief account of the quantum behaviour of photons. This is a rich field for practical physics and learners will have many opportunities to extend their experimental and analytical skills. In addition, the topics provide a picture of the development of theories and understanding over time (HSW1, HSW2, HSW7). Quantum behaviour is discussed through considering possible photon paths, avoiding the wave/particle dichotomy.